Lab – RESTCONF with Postman

Objectives

Part 1: Setup HTTP Headers in Postman Carlos Antonio Frias Juárez

Background / Scenario

RESTCONF is a RESTful API interface that provides a programmatic interface for accessing data defined in YANG device models.

In this lab, you will learn how to interact with the RESTCONF interface using the Postman application to retrieve the device's configuration, update and create new interfaces and retrieve operation data.

Required Resources

- Postman
- Access to a router with the IOS XE operating system version 16.6 or higher.

Instructions

Part 1: Setup HTTP headers and Authentication in Postman

In this part, you will setup the HTTP Headers in Postman to work with RESTCONF REST API interfaces.

Step 1: Configure Postman.

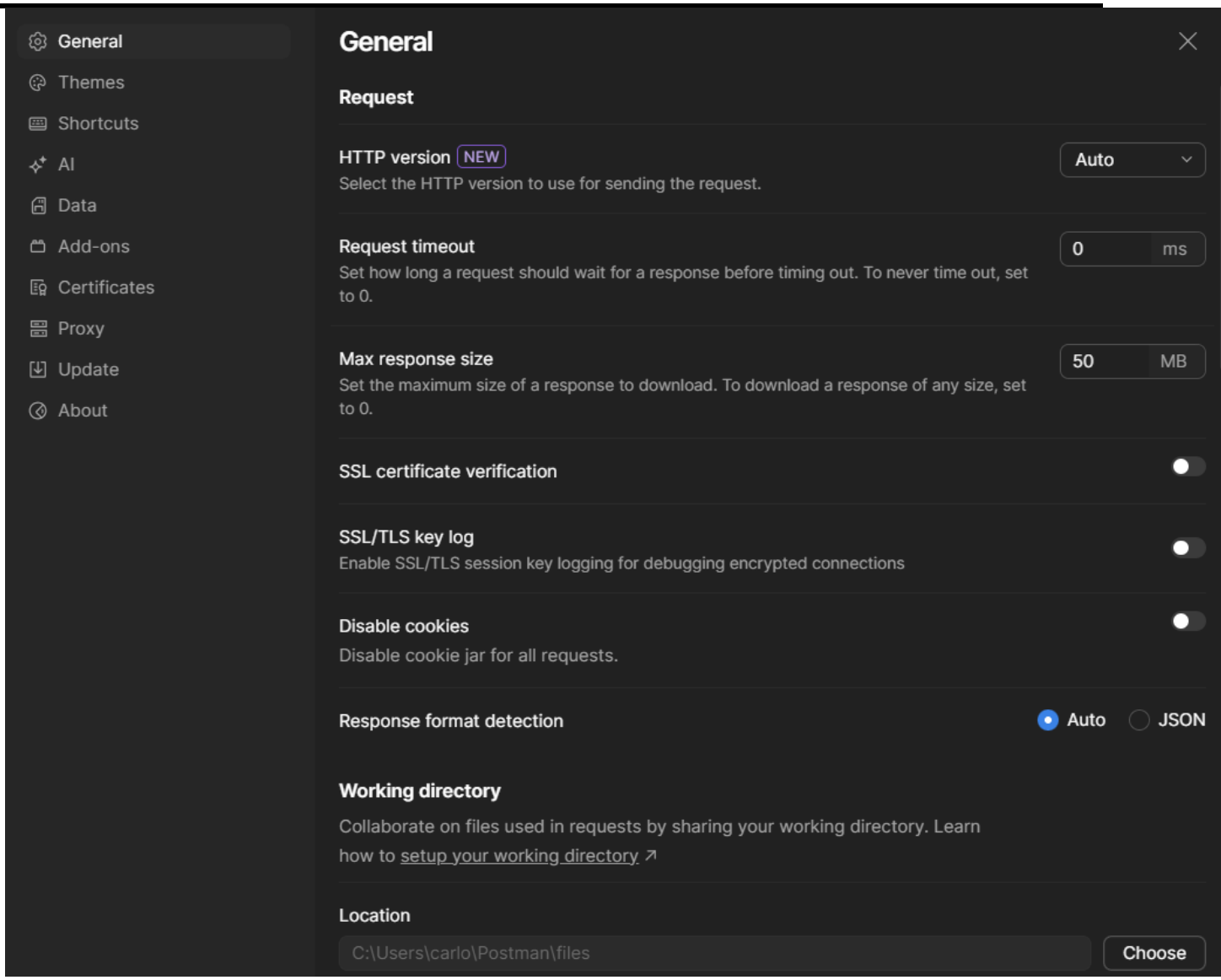
Configure a Postman setting.

a. Open Postman and create a new tab if necessary.

b. Click **File > Settings**.

c. Under the **General** tab, set the **SSL certificate verification** to **OFF**. Close the **Settings** dialog box.

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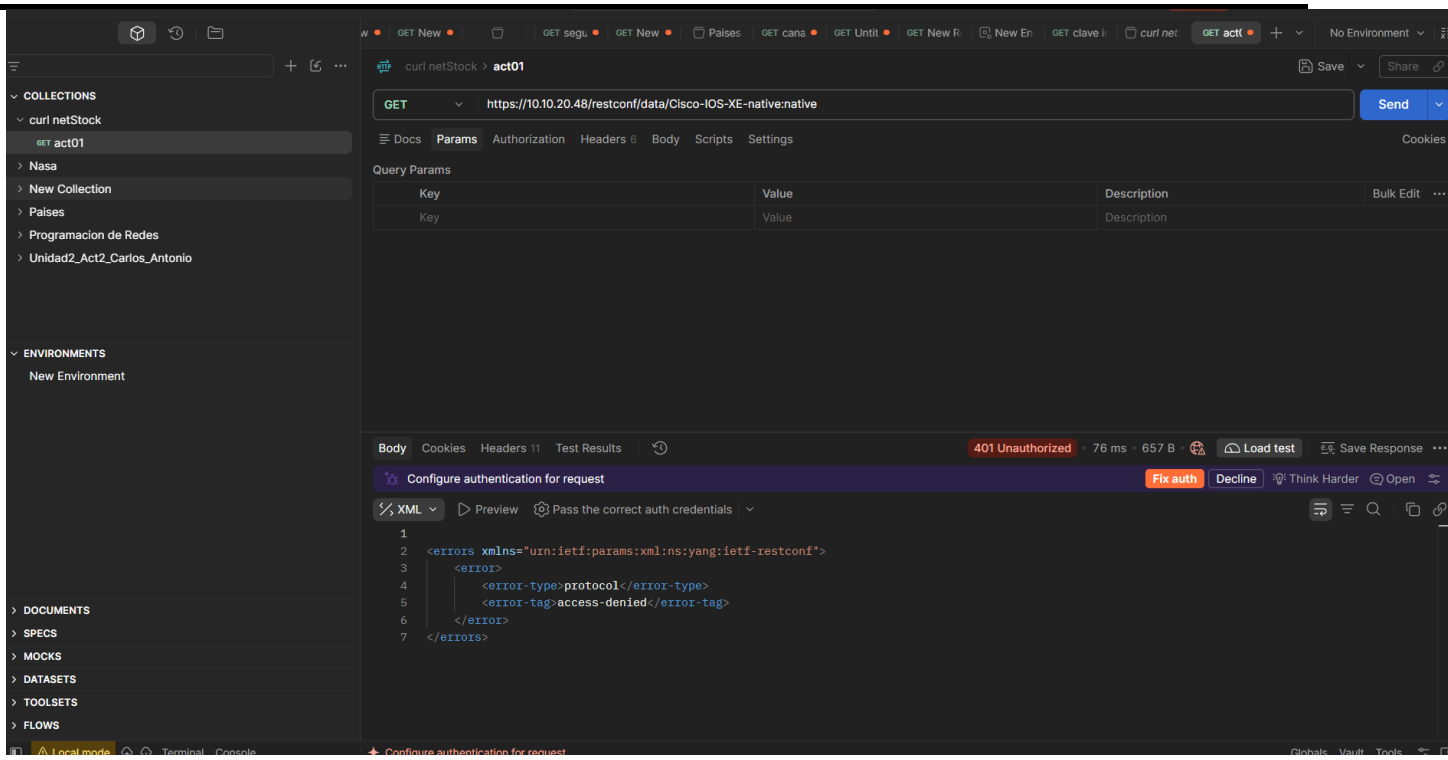


Step 2: Select the method and enter the required URL.

a. Next to the URL box, select the request method to be **GET**.

b. Enter the URL for API endpoint: https://10.10.20.48/restconf/data/Cisco-IOS-XE-native:native (adjust the IP address to match the router's current address)

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Step 3: Enter the authentication header information.

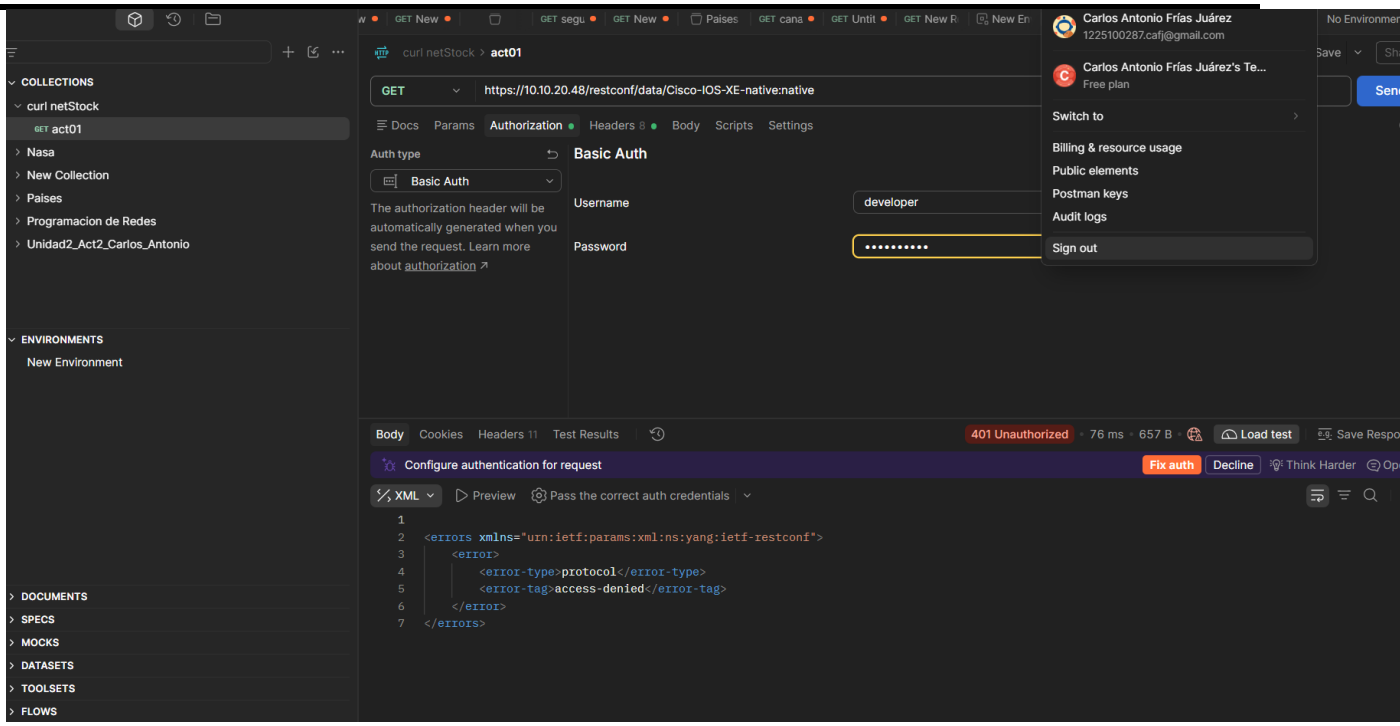
a. Below the URL input field, select **Authorization**.

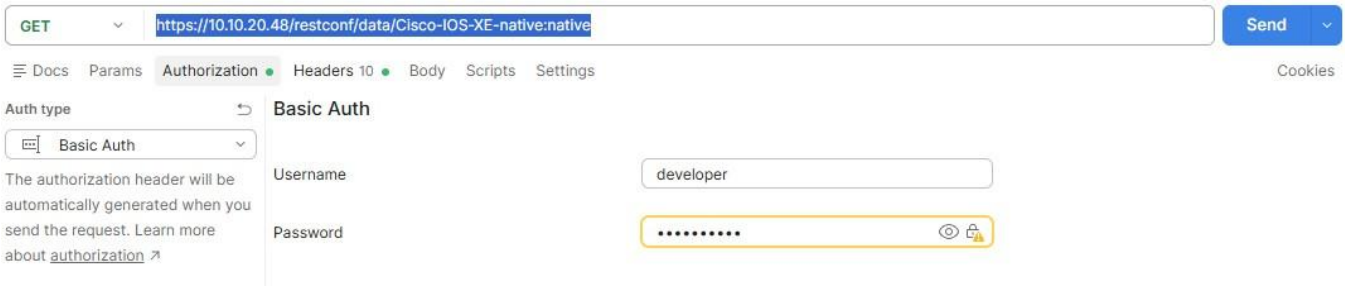
b. Select the type **Basic Auth** and enter the username and password for authenticating to the RESTCONF API service:
`developer`

C1sco12345

c. Click on the **Preview Request** to add the Authentication header to the Headers.

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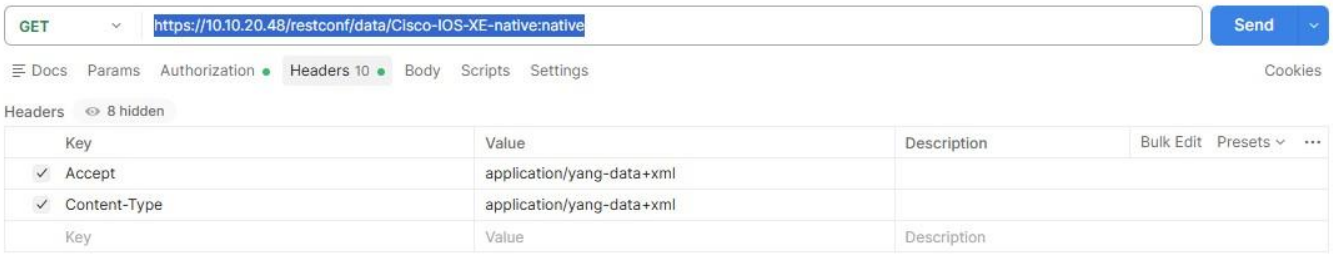
Step 4: Enter the additional Headers.

a. Next, below the URL input field, select **Headers**.

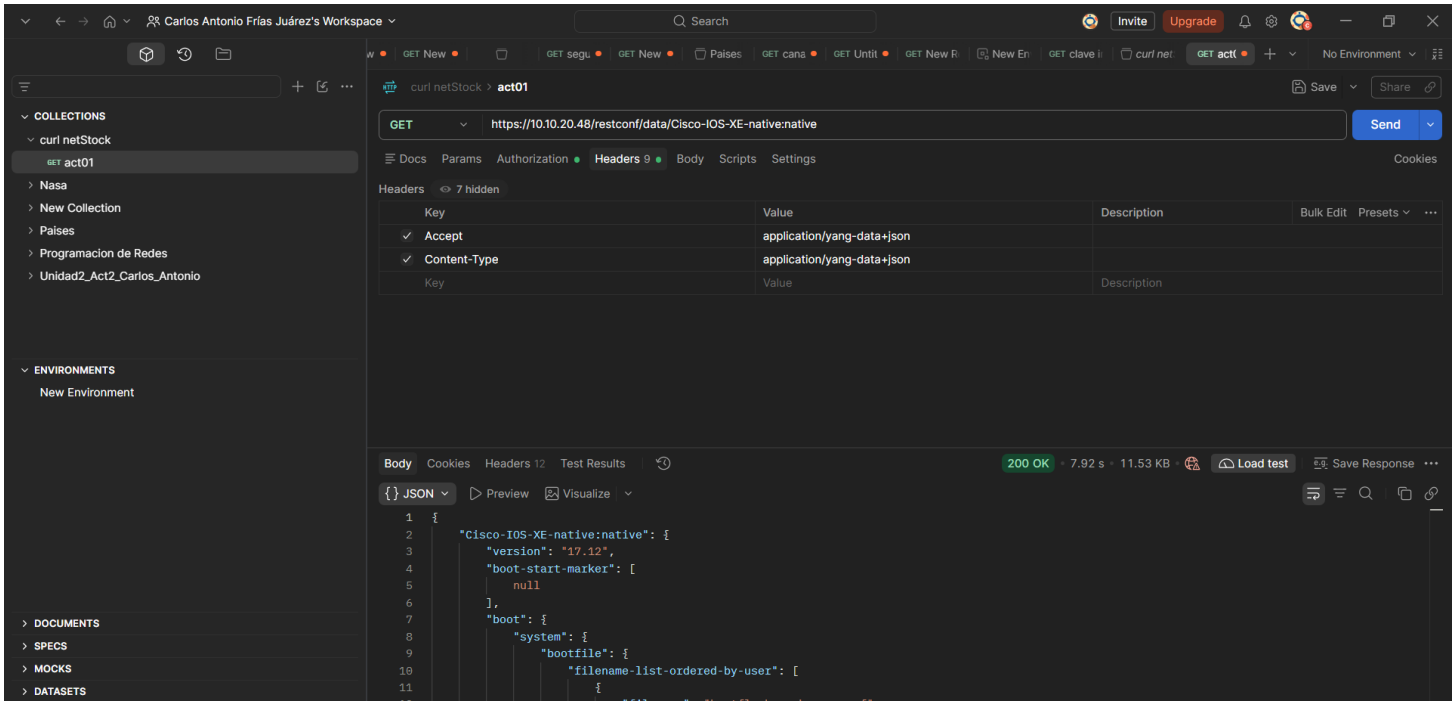
b. Under **Headers**, click the **New key** field under the **Key** column and enter **Content-Type** to define what is the encoding of data that are sent with Postman in the REST API request.

c. Next again under **Headers**, click the **New key** field under the **Key** column and enter **Accept** to define in what encoding you wish to receive back the data replies from the REST API request.

d. In the value columns, enter `application/yang-data+json` when sending or receiving data that are in JSON encoding. When sending or receiving data in XML encoding (default), use `application/yangdata+xml`.



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Step 5: Send the request.

a. Click **Send**.

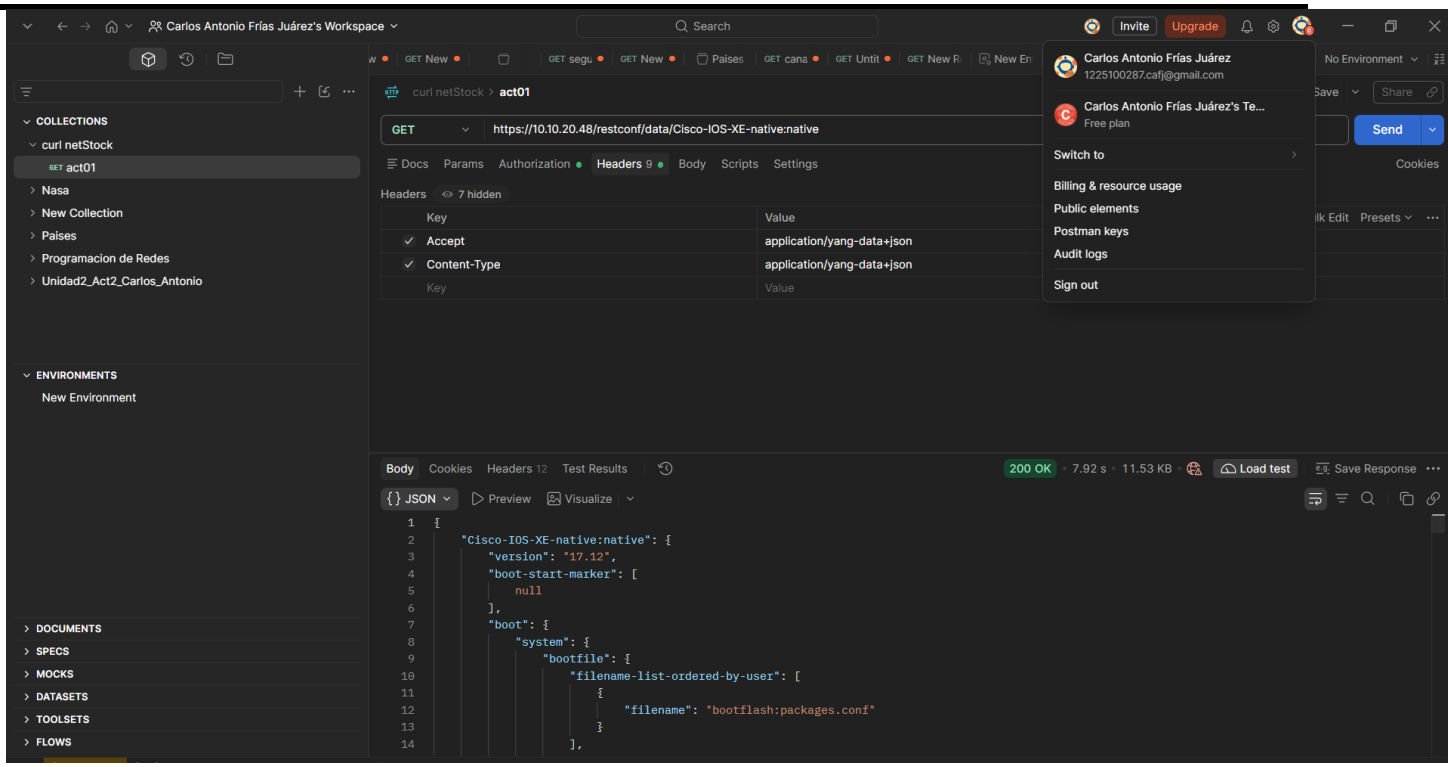
b. After a brief delay, you should see the response appear below the request information. Scroll down, if necessary to view the response data.

c. When the **Accept** header was set to `application/yang-data+json` the response consists of JSON encoded data.

d. Uncheck the **Accept** header, or change it to `application/yang-data+xml` (default behavior when no Accept header is defined) and click on **Send**. Now the response consists of XML encoded data:

Note: If there is an error, check the Status of the request. Check the meaning of the status code. 200 means success. A 404 error may mean that the URL was entered incorrectly. A 401 or 403 error could indicate a problem with the authentication, so check that the credentials are entered correctly.

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Part 2: Retrieve Interface Configuration

Step 1: Retrieve YANG data in XML.

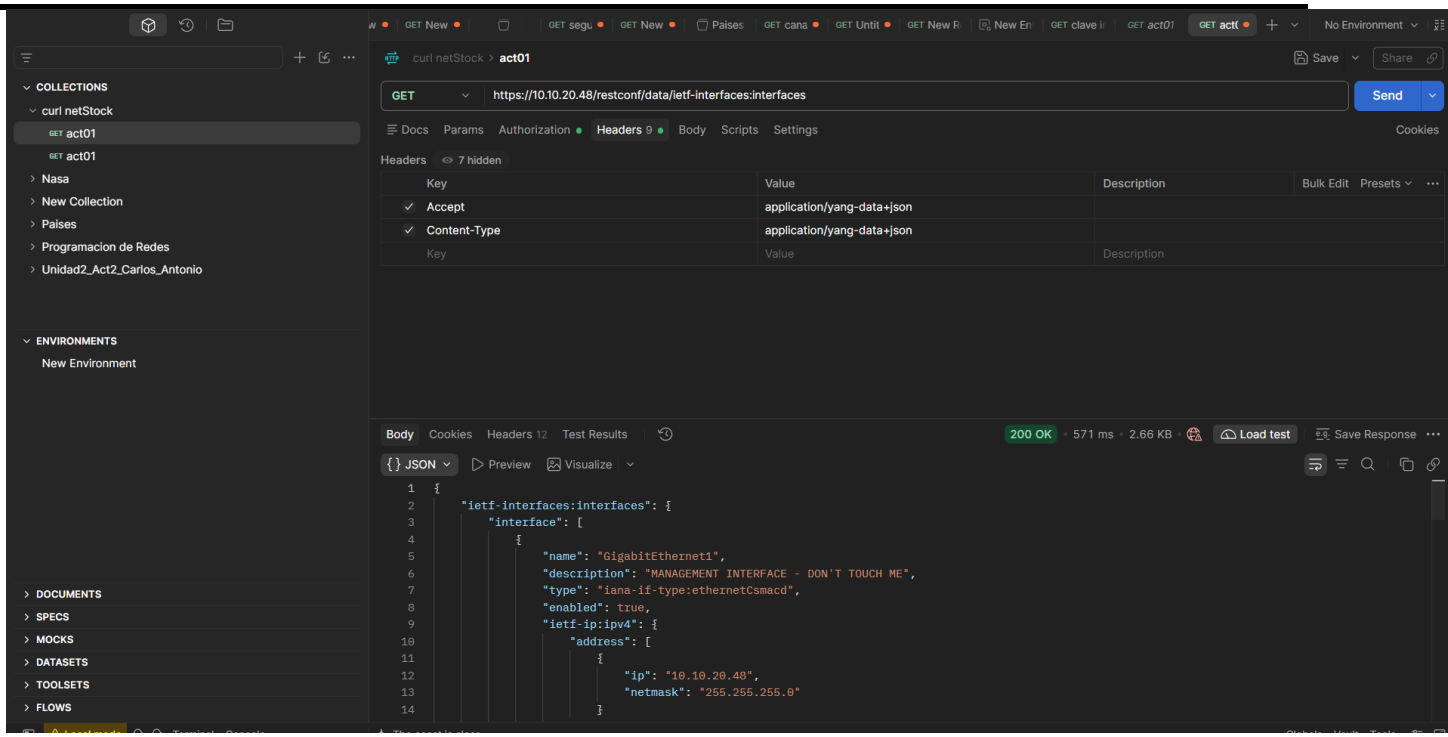
a. Duplicate the existing REST API request in Postman by right clicking on the request and selecting Duplicate Tab to create a new request with prefilled existing Headers:

b. In the URL for the API endpoint, enter after the host and port portion: `/restconf/data/ietf-interfaces:interfaces` to retrieve data defined in the **ietf-interfaces** YANG model in the **interfaces** container.

c. Click **Send** to send the RESTCONF API request. With unchecked **Accept** header or the **Accept** header set to `application/yang-data+xml`, the result YANG data are returned in XML encoding.

d. Check the Accept header and set it to `application/yang-data+json` to receive YANG data formatted in JSON encoding.

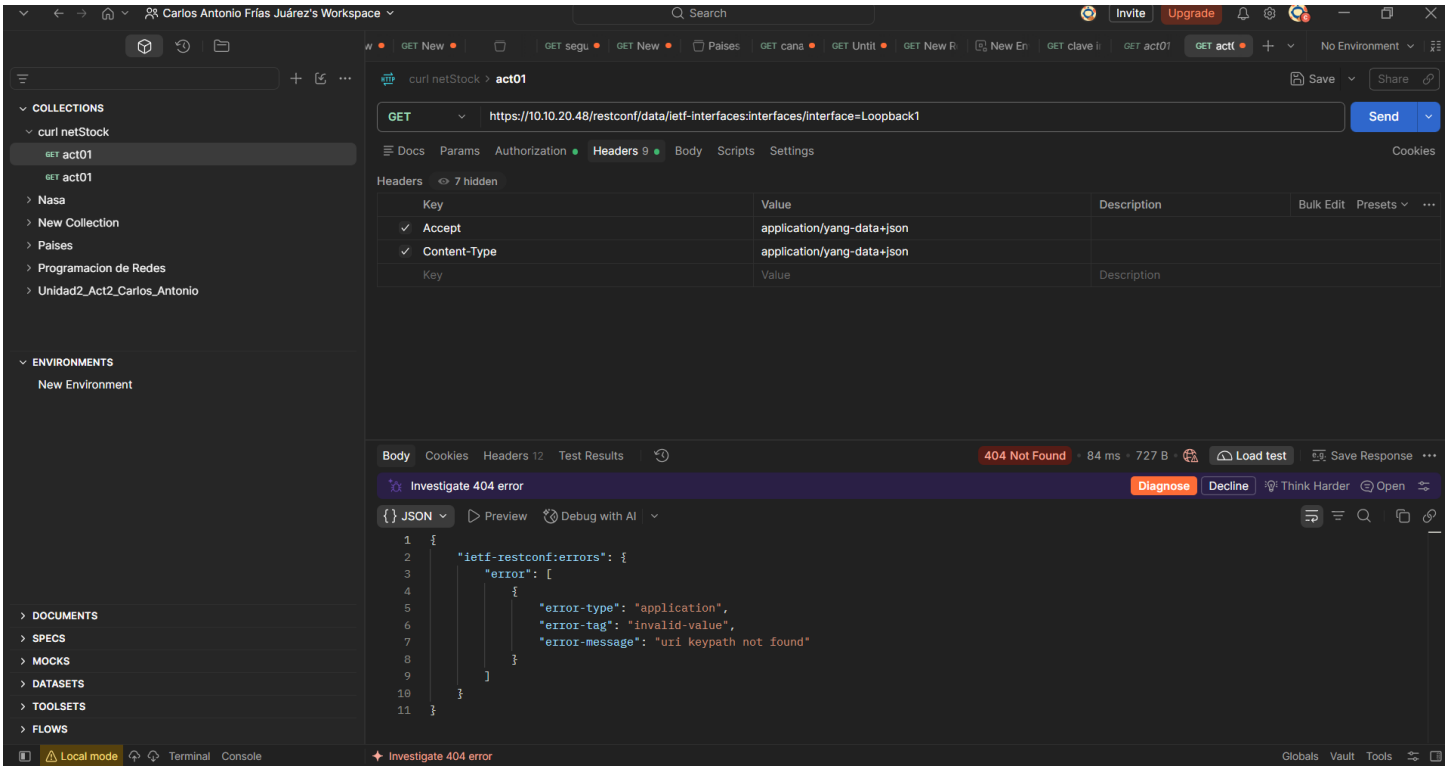
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Step 2: Retrieve information about a single interface

a. In the URL for the API endpoint, enter after the host and port portion: `/restconf/data/ietf-interfaces:interfaces/interface=Loopback1` to retrieve data for the **Loopback1** interface only.

b. Click **Send** …

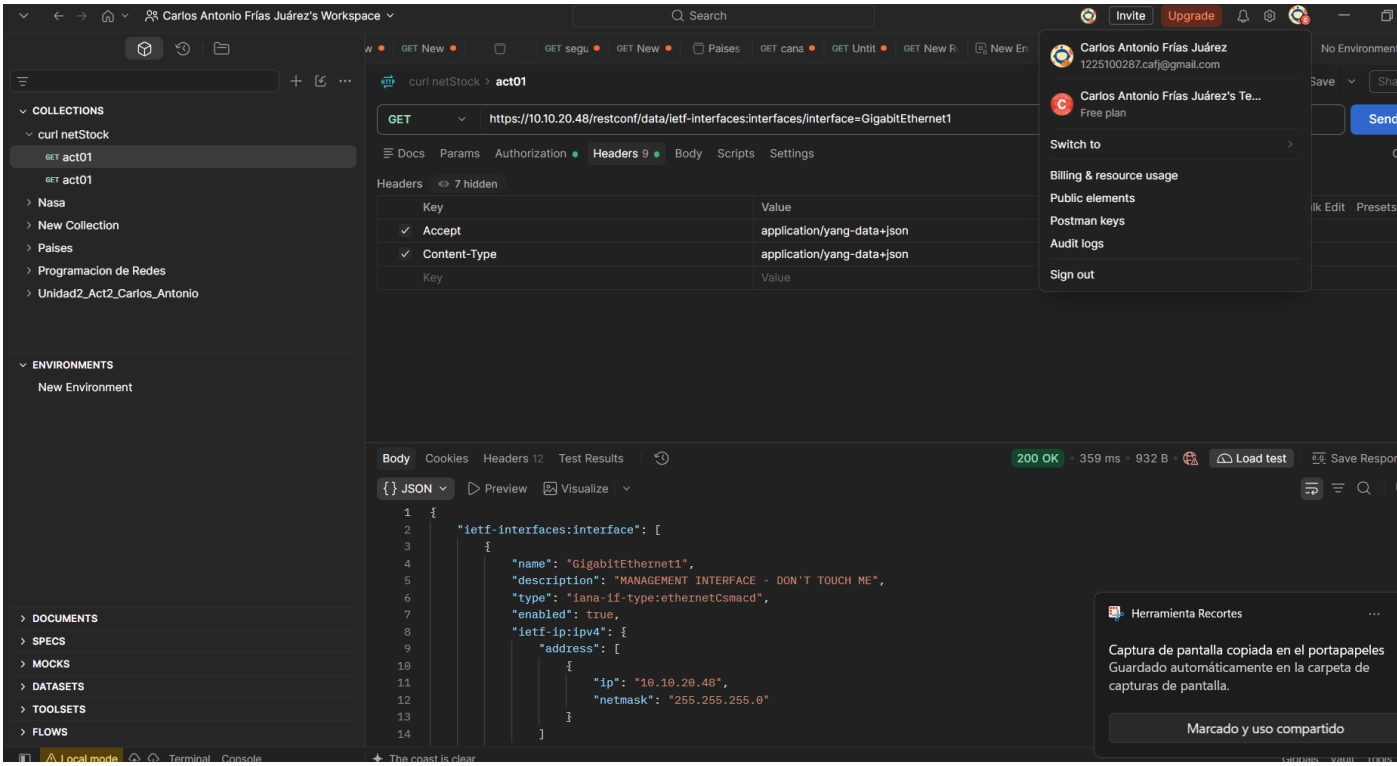


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3: Retrieve operation data

a. Change the RESTCONF API Endpoint to retrieve operation and statistical data about the interfaces. You can use the “pyang” tool to discover what container in the **ietf-interfaces** YANG model includes the operational and stats data.

b. What is the new URL? /restconf/data/ietf-interfaces:interfaces-state



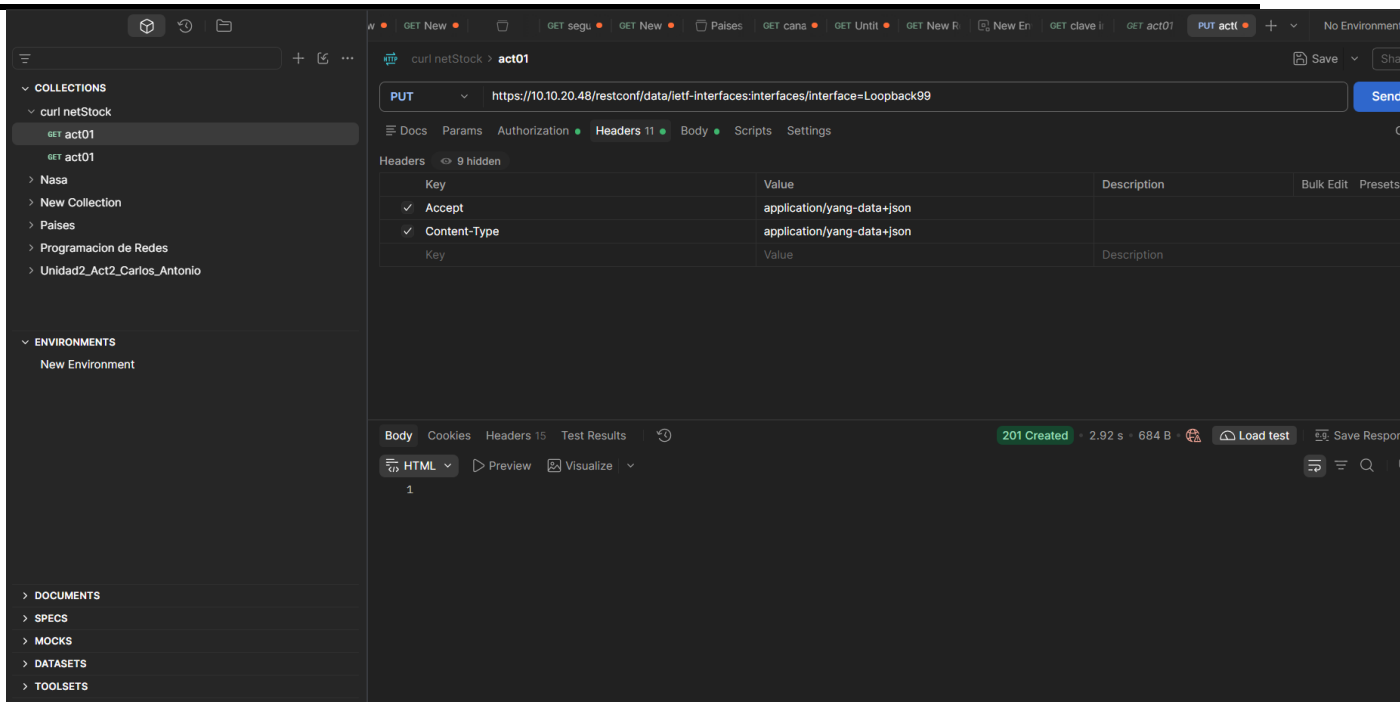
Part 3: Update Interface Configuration

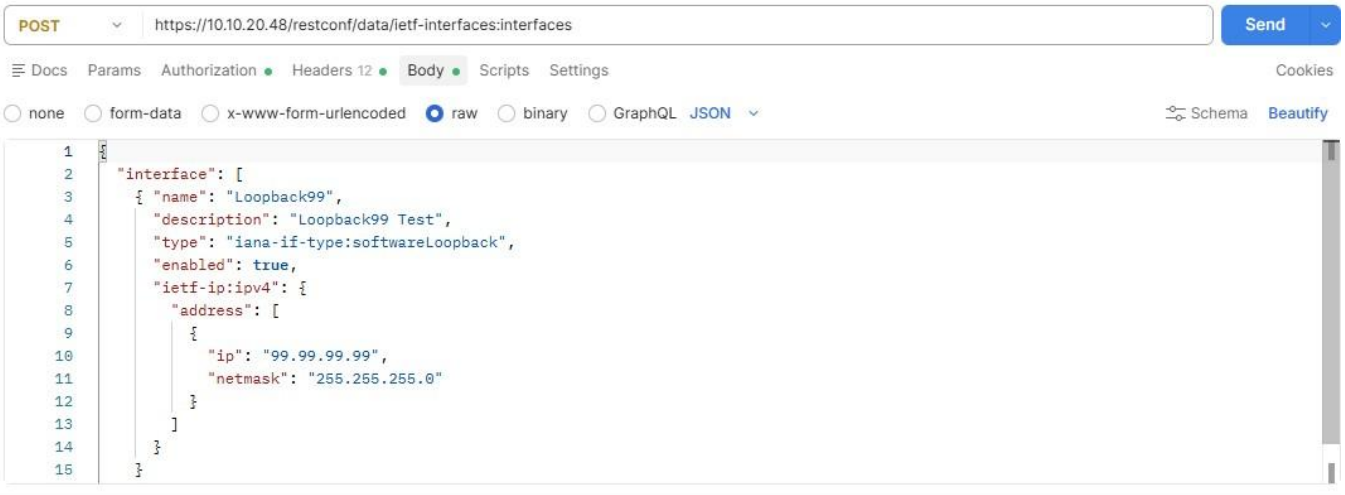
a. Duplicate the existing REST API request in Postman by right clicking on the request and selecting Duplicate Tab to create a new request with prefilled existing Headers.

b. In the URL for the API endpoint, change the method from **GET** to **PUT** and enter after the host and port portion: `/restconf/data/ietf-interfaces:interfaces/interface=Loopback99` to create a new interface **Loopback99**.

c. In the Body part of the request, enter the YANG data you want to apply to the device’s configuration (copy, paste and customize the JSON data from the previous GET request on Loopback1 – change the interface name and IP address).

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d. Click **Send** to create a new interface Loopback99 with the IP address 99.99.99.99/24.

e. In the Output (scroll down), you should see the HTTP Error code **201 Created** indicating that the new interface has been created:

f. Verify using the IOS CLI that the new Loopback99 interface has been created (sh ip int brief).

g. Update the RESTCONF API request to delete the interface **Loopback99**.

h. What changes were needed to be made in the RESTCONF request to delete the interface? Para eliminar la interfaz solo fue necesario cambiar el método HTTP de PUT a DELETE

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```
S C:\Users\carlo> ssh developer@10.10.20.48
developer@10.10.20.48) Password:

Welcome to the DevNet Sandbox for Cat8000V and IOS XE

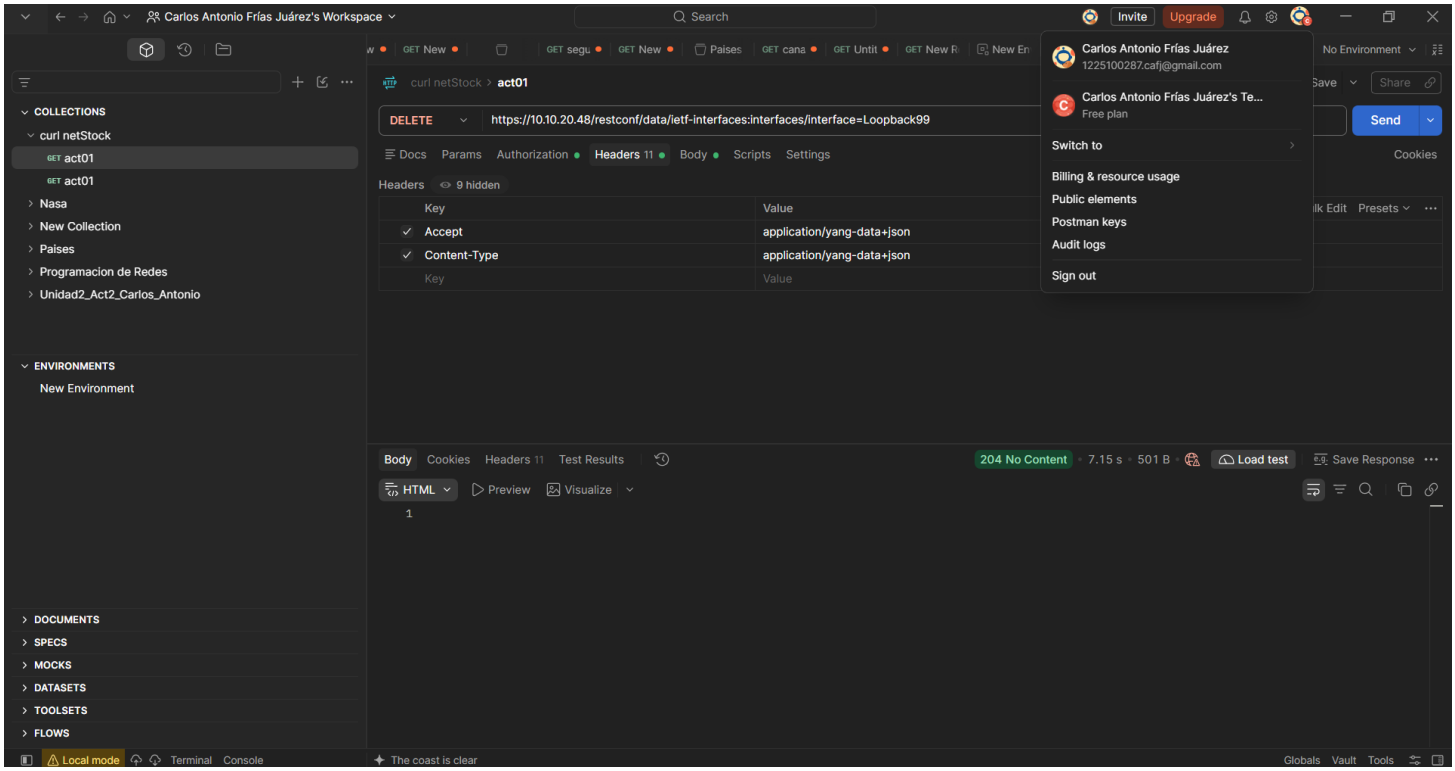
The following programmability features are already enabled:

-NETCONF
-RESTCONF

Thanks for stopping by.



at8000v#show
at8000v#show ip interface brief
nterface              IP-Address      OK? Method Status                Protocol
igabitEthernet1       10.10.20.48     YES NVRAM  up                    up
igabitEthernet2       unassigned      YES NVRAM  administratively down down
igabitEthernet3       unassigned      YES NVRAM  administratively down down
oopback0              10.0.0.1        YES NVRAM  up                    up
oopback10             unassigned      YES unset  up                    up
oopback99             99.99.99.99     YES other  up                    up
oopback109            10.255.255.9    YES NVRAM  up                    up
irtualPortGroup0      192.168.1.1     YES NVRAM  up                    up
at8000v#
```



DESAFÍOS

Enviar la solicitud GET para obtener la configuración global del router (/restconf/data/Cisco-IOS-XEnative:native). Deben hacer una captura donde se observe la respuesta formateada en JSON, y otra donde se observe en XML alterando únicamente los encabezados (Headers).

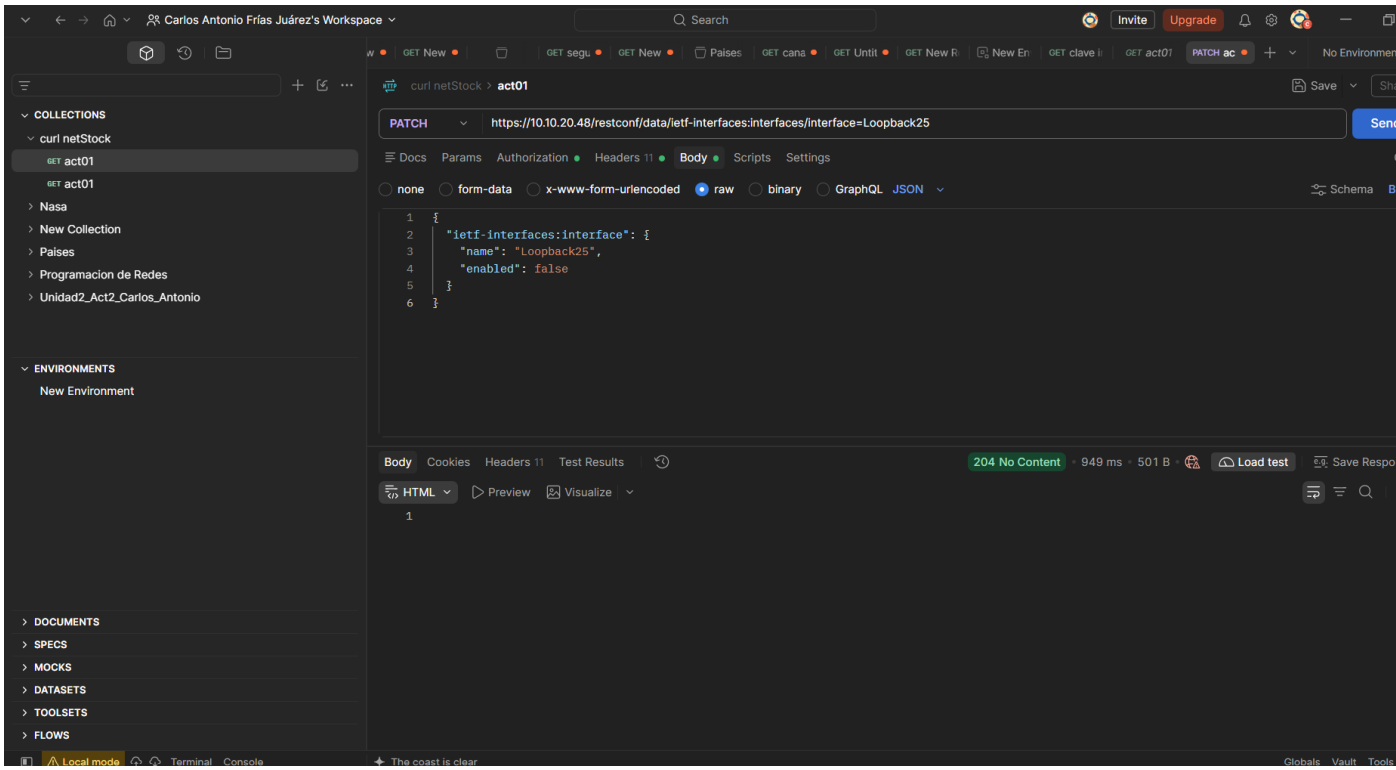
En lugar de solicitar todas las interfaces (/ietf-interfaces:interfaces), debes modificar la URI para consultar únicamente la interfaz GigabitEthernet1 usando JSON.

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Crear una nueva interfaz Loopback utilizando su número de lista o las últimas dos cifras de su matrícula como el ID de la interfaz (ej. Loopback25) y asignarle la IP. Debes obtener un código de estado 201 Created en Postman e incrustar la descripción "Interfaz creada por [Tú Nombre]"

Cambiar el estado de la interfaz recién creada a "deshabilitada" ("enabled": false) o modifica únicamente la description utilizando el método PATCH en lugar de PUT.

Eliminar la interfaz creada en el Desafío 3 utilizando el método **DELETE.**



Preguntas de Reflexión

¿Qué encabezado HTTP le indica al servidor en qué formato queremos recibir la información?

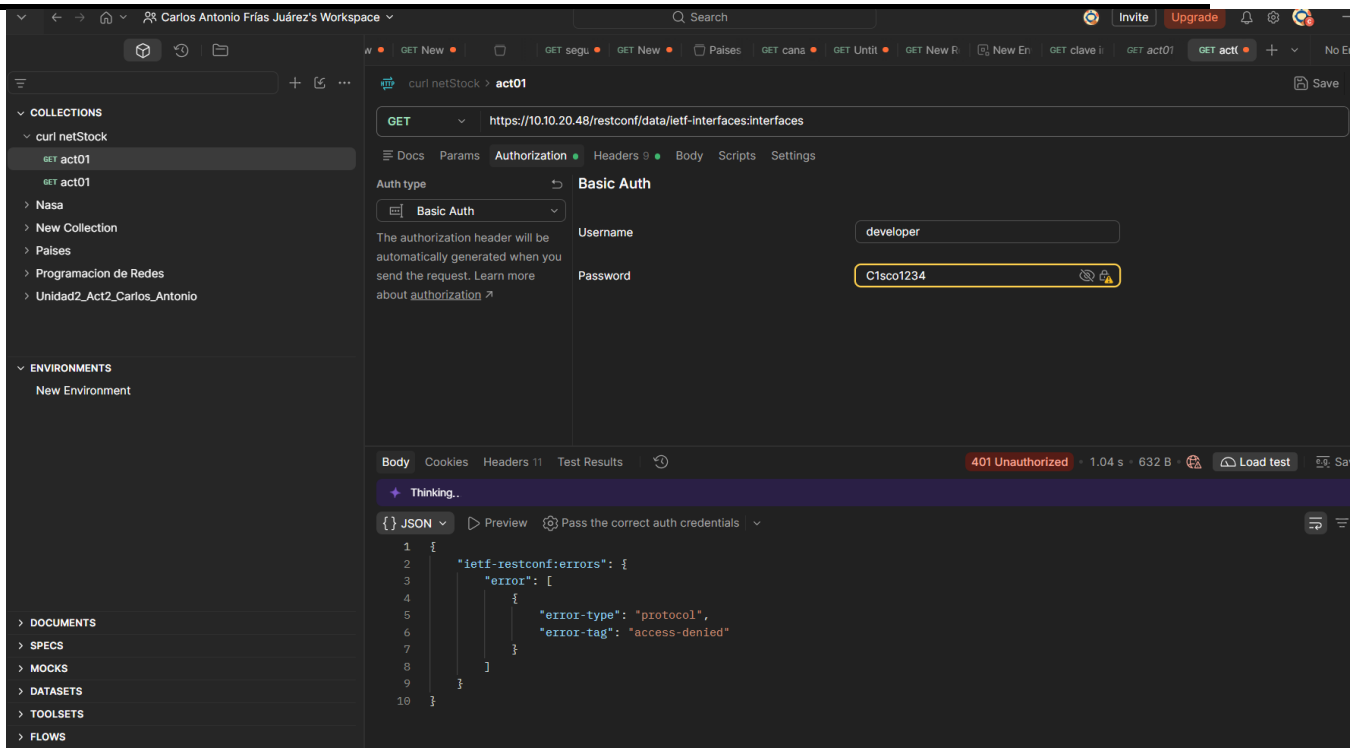
El encabezado Accept. Con application/yang-data+json el router responde en JSON y con application/yang-data+xml responde en XML. (Content-Type es distinto: indica el formato de los datos que enviamos en el cuerpo.)

¿Qué código de estado HTTP devuelve el router tras eliminar exitosamente la interfaz?

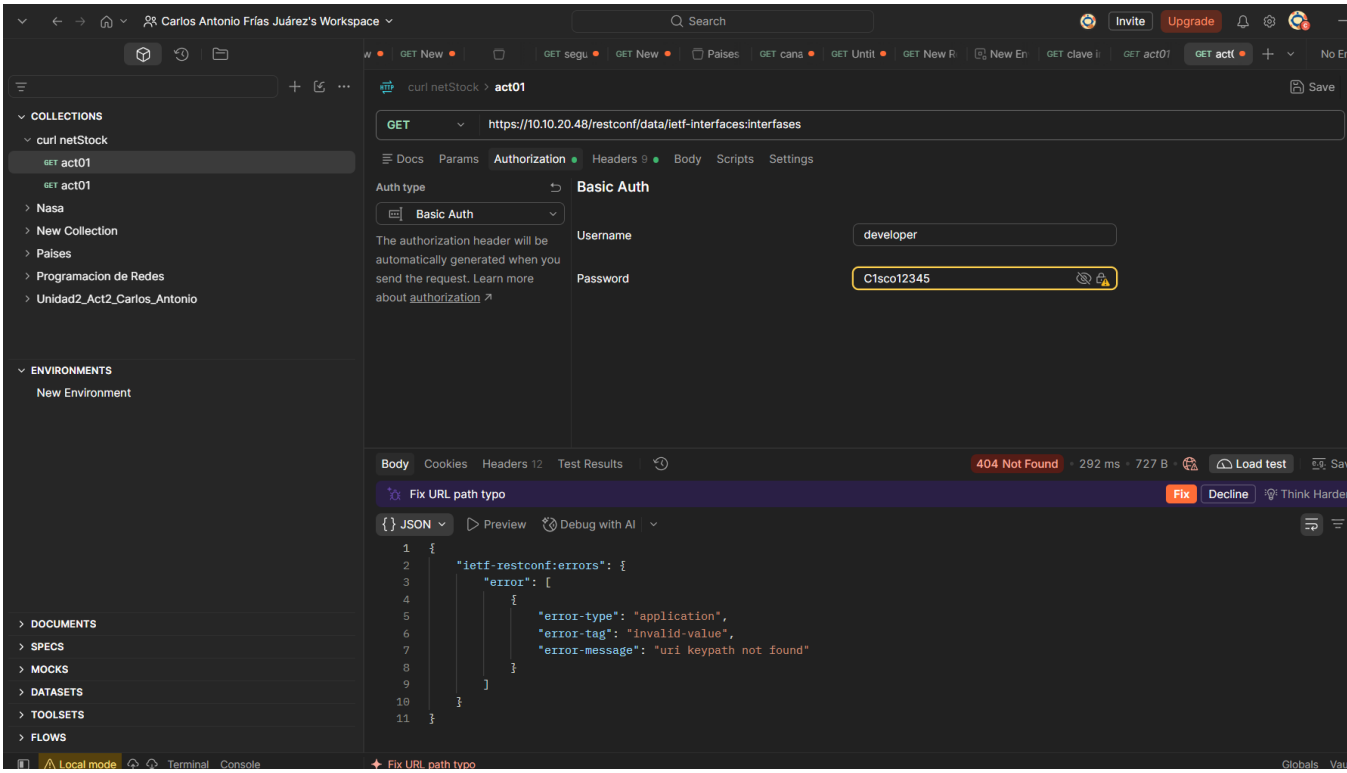
204 No Content: la operación fue exitosa y la respuesta viene sin cuerpo.

Escribir mal las credenciales en Basic Auth (Ej. intentar con developer / Cisco123) -> Esperado: 401 Unauthorized.

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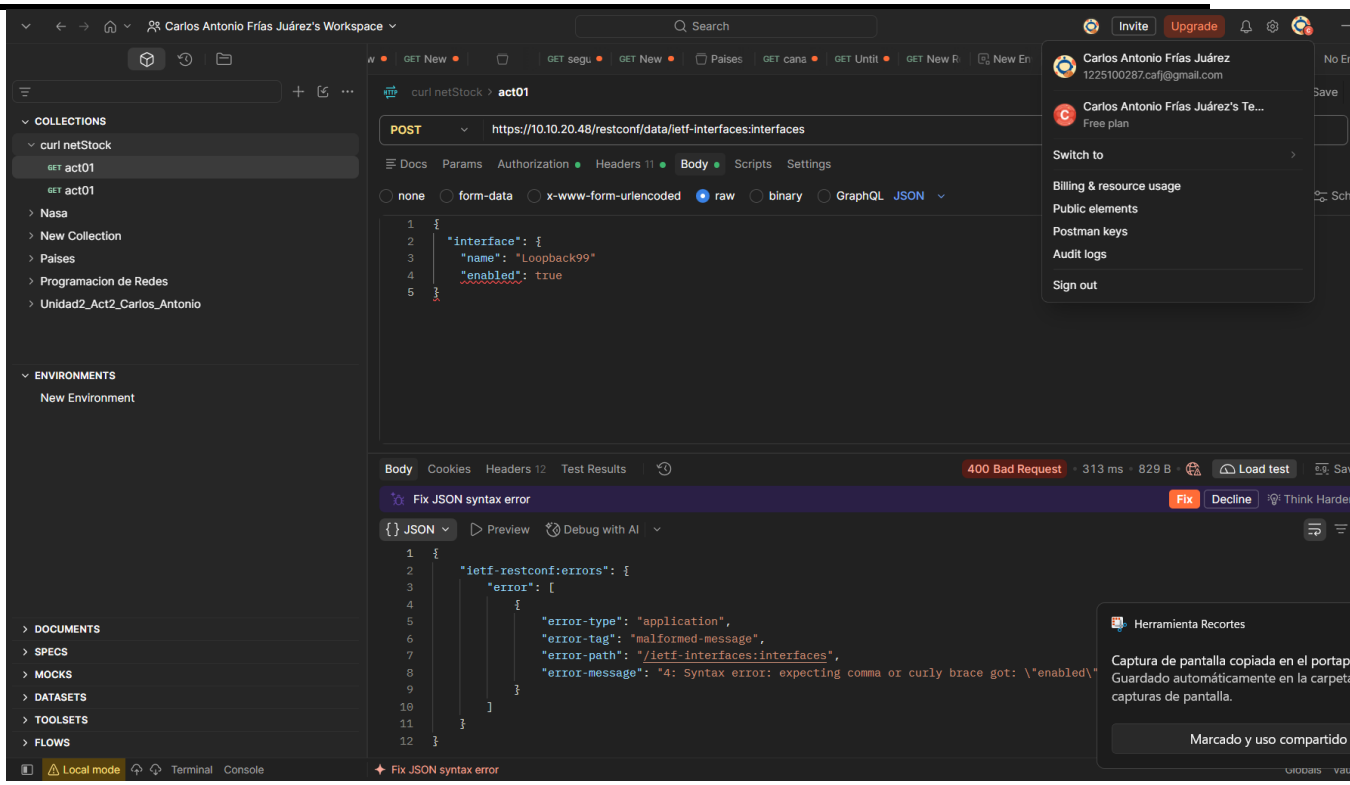
Escribir mal la URL (Ej. /restconf/data/ietf-interfaces:interfases) -> Esperado: 404 Not Found.



Enviar un cuerpo JSON mal formado o con sintaxis inválida en un POST -> Esperado: 400 Bad Request o 415 Unsupported Media Type.

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Lab – RESTCONF with Postman



Lista de Cotejo: Evaluación de Práctica RESTCONF y Entrega en GitHub

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Criterio / Indicador	Puntos Posibles	Puntos Obtenidos	Observaciones
1. Práctica Guiada (RESTCONF Básico)	**2.5 pts**		
* Configuración correcta de la autenticación desactivación de SSL en Postman.	0.5		
* Configuración correcta de los encabez Content-Type) para JSON y XML.	1.0		
* Obtención exitosa de respuestas HTTP 2 del dispositivo (native e interfaces).	1.0		
2. Resolución de Desafíos Prácticos	**4.0 pts**		
* **Creación de Recurso:** Creación correcta mediante solicitud POST/PUT obteniendo Created.	1.5		
* **Modificación:** Modificación exitosa del est interfaz mediante PUT o PATCH.	1.0		
* **Eliminación y Depuración:** Eliminación d DELETE y correcta interpretación/provocaci prueba (ej. 401, 404).	1.5		
3. Preguntas de Reflexión y Análisis	**2.0 pts**		
* Explica con precisión la función del encabe diferencia en la respuesta al cambiar en	1.0		
* Explica la diferencia práctica entre los métodos HTTP utilizados (GET, POST, PUT, DELETE) y el significado de los códigos de estado obtenidos (200, 201, 401, 404).	1.0		
4. Entrega y Flujo en GitHub	**1.5 pts**		
* El Pull Request fue enviado correctamente el nombre exacto: feat/unidad4_act01.	0.5		
* La entrega incluye las capturas de pantalla colección exportada de Postman dentro del	0.5		
* El Pull Request incluye un título claro y una trabajo realizado.	0.5		
TOTAL	**10.0 pts**		

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